

Research & Development / AI Assignment

Parametric Curve Parameter Estimation

Nonlinear Parameter Estimation from Observed Two-Dimensional Data

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Technologies Used

Python NumPy Pandas SciPy Matplotlib

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1 Introduction

This project addresses a nonlinear inverse parameter estimation problem. The objective is to recover the unknown parameters of a two-dimensional parametric curve from a set of observed (x, y) points.

The given curve is described by the following equations:

$$x(t) = t \cos(\theta) - e^{M|t|} \sin(0.3t) \sin(\theta) + X \quad (1)$$

$$y(t) = 42 + t \sin(\theta) + e^{M|t|} \sin(0.3t) \cos(\theta) \quad (2)$$

The unknown parameters are:

$$\theta, \quad M, \quad X \quad (3)$$

The parameter θ controls the orientation of the curve, M controls the exponential growth or decay of the oscillatory component, and X controls the horizontal translation of the curve.

The goal is to estimate these parameters such that the reconstructed curve matches the supplied observations as closely as possible.

2 Problem Statement

Given a dataset containing observed two-dimensional points, determine the values of θ , M , and X that best explain the observations.

The parameter constraints specified by the problem are:

$$0^\circ < \theta < 50^\circ \quad (4)$$

$$-0.05 < M < 0.05 \quad (5)$$

$$0 < X < 100 \quad (6)$$

The hidden curve parameter t is expected to satisfy:

$$6 < t < 60 \quad (7)$$

The quality of the fitted solution is evaluated by comparing the reconstructed points with the supplied observations.

3 Dataset

The input dataset is stored in:

`data/xy_data.csv`

The dataset contains 1,500 two-dimensional observations.
Each observation consists of:

x
 y

The observed points are assumed to have been generated from the given parametric curve, possibly with a small amount of numerical or measurement error.

The objective is therefore to identify the parameter values that produce the closest possible representation of these observations.

4 Mathematical Formulation

Because the supplied range of t is positive,

$$t > 0 \tag{8}$$

and therefore:

$$|t| = t \tag{9}$$

The original equations can consequently be written as:

$$x(t) = t \cos(\theta) - e^{Mt} \sin(0.3t) \sin(\theta) + X \tag{10}$$

$$y(t) = 42 + t \sin(\theta) + e^{Mt} \sin(0.3t) \cos(\theta) \tag{11}$$

For convenience, define the oscillatory component:

$$A = e^{Mt} \sin(0.3t) \tag{12}$$

The equations then become:

$$x - X = t \cos(\theta) - A \sin(\theta) \tag{13}$$

$$y - 42 = t \sin(\theta) + A \cos(\theta) \tag{14}$$

This representation shows that the observed coordinates can be interpreted as a rotation of the two components t and A .

4.1 Recovery of the Hidden Parameter t

To recover t , multiply the first equation by $\cos(\theta)$ and the second equation by $\sin(\theta)$:

$$(x - X) \cos(\theta) = t \cos^2(\theta) - A \sin(\theta) \cos(\theta) \tag{15}$$

and

$$(y - 42) \sin(\theta) = t \sin^2(\theta) + A \sin(\theta) \cos(\theta) \tag{16}$$

Adding the two equations gives:

$$(x - X) \cos(\theta) + (y - 42) \sin(\theta) = t [\cos^2(\theta) + \sin^2(\theta)] \quad (17)$$

Using the identity

$$\cos^2(\theta) + \sin^2(\theta) = 1 \quad (18)$$

we obtain:

$$\boxed{t = (x - X) \cos(\theta) + (y - 42) \sin(\theta)} \quad (19)$$

Thus, for any candidate values of θ and X , the corresponding value of t can be recovered directly from each observed point.

4.2 Recovery of the Oscillatory Component

A perpendicular projection can be used to recover A .

Multiplying the first equation by $-\sin(\theta)$ and the second equation by $\cos(\theta)$ gives:

$$-(x - X) \sin(\theta) + (y - 42) \cos(\theta) = A \quad (20)$$

Since:

$$A = e^{Mt} \sin(0.3t) \quad (21)$$

the transformed model becomes:

$$\boxed{-(x - X) \sin(\theta) + (y - 42) \cos(\theta) = e^{Mt} \sin(0.3t)} \quad (22)$$

This transformation is useful because it eliminates the explicit need to optimize over every individual value of t .

5 Parameter Estimation Method

Let the unknown parameter vector be:

$$\mathbf{p} = [\theta, M, X] \quad (23)$$

For every observed point (x_i, y_i) , the hidden parameter t_i is calculated as:

$$t_i = (x_i - X) \cos(\theta) + (y_i - 42) \sin(\theta) \quad (24)$$

The observed transformed oscillatory component is:

$$v_i = -(x_i - X) \sin(\theta) + (y_i - 42) \cos(\theta) \quad (25)$$

The model prediction is:

$$\hat{v}_i = e^{Mt_i} \sin(0.3t_i) \quad (26)$$

Therefore, the residual associated with the i -th point is:

$$r_i = v_i - \hat{v}_i \quad (27)$$

The optimization problem is:

$$\min_{\theta, M, X} \sum_i r_i^2 \quad (28)$$

subject to:

$$0 < \theta < 50 \quad (29)$$

$$-0.05 < M < 0.05 \quad (30)$$

$$0 < X < 100 \quad (31)$$

This is a bounded nonlinear least-squares problem.

6 Optimization Algorithm

The implementation uses the `scipy.optimize.least_squares` algorithm.

The optimization procedure can be summarized as follows:

1. Load the observed x and y coordinates.
2. Select an initial estimate for θ , M , and X .
3. Convert θ from degrees to radians for numerical computation.
4. Recover the corresponding t_i values for every observed point.
5. Calculate the transformed oscillatory component.
6. Calculate the model prediction $e^{Mt_i} \sin(0.3t_i)$.
7. Compute the residual between the observed and predicted oscillatory components.
8. Minimize the residual using bounded nonlinear least squares.
9. Substitute the estimated parameters into the original parametric equations.
10. Reconstruct the complete curve.
11. Evaluate the reconstruction using the L1 error.

7 Initial Values and Parameter Bounds

The optimizer is initialized using values inside the allowed parameter region.

Parameter	Initial Value	Allowed Range
θ	25°	$0^\circ < \theta < 50^\circ$
M	0	$-0.05 < M < 0.05$
X	50	$0 < X < 100$

Table 1: Initial Values and Parameter Constraints

The use of bounds prevents the optimization procedure from exploring physically or mathematically invalid parameter values.

8 Implementation

The core residual function is implemented in Python as follows.

```

1 def residuals(params):
2
3     theta_deg, M, X = params
4
5     theta = np.deg2rad(theta_deg)
6
7     cos_theta = np.cos(theta)
8     sin_theta = np.sin(theta)
9
10    # Recover hidden parameter t
11    t = (
12        (x - X) * cos_theta
13        + (y - 42) * sin_theta
14    )
15
16    # Recover observed oscillatory component
17    v = (
18        -(x - X) * sin_theta
19        + (y - 42) * cos_theta
20    )
21
22    # Model prediction
23    v_pred = (
24        np.exp(M * t)
25        * np.sin(0.3 * t)
26    )
27
28    # Residual
29    return v - v_pred
    
```

The bounded optimizer is applied using:

```

1 result = least_squares(
2     residuals,
3     initial_guess,
4     bounds=(lower_bounds, upper_bounds),
5     max_nfev=100000,
6     xtol=1e-14,
7     ftol=1e-14,
8     gtol=1e-14
9 )
    
```

The use of strict convergence tolerances allows the optimization procedure to continue until the numerical solution has sufficiently converged.

9 Estimated Parameters

The optimization produces the following parameter estimates:

Parameter	Estimated Value
θ	30.0000°
M	0.0300
X	55.0000

Table 2: Final Estimated Parameters

Therefore, the final estimated parameters are:

$$\boxed{\theta = 30^\circ, \quad M = 0.03, \quad X = 55} \quad (32)$$

All three estimated values satisfy the parameter constraints specified in the problem statement.

10 Curve Reconstruction

Substituting the estimated parameters into the original equations gives:

$$x(t) = t \cos(30^\circ) - e^{0.03t} \sin(0.3t) \sin(30^\circ) + 55 \quad (33)$$

and:

$$y(t) = 42 + t \sin(30^\circ) + e^{0.03t} \sin(0.3t) \cos(30^\circ) \quad (34)$$

Using:

$$\sin(30^\circ) = 0.5 \quad (35)$$

and:

$$\cos(30^\circ) \approx 0.8660254 \quad (36)$$

the final curve can be written as:

$$x(t) = 0.8660254t - 0.5e^{0.03t} \sin(0.3t) + 55 \quad (37)$$

$$y(t) = 42 + 0.5t + 0.8660254e^{0.03t} \sin(0.3t) \quad (38)$$

These equations represent the reconstructed curve obtained from the estimated parameters.

11 Error Evaluation

To evaluate the quality of the reconstructed curve, the predicted coordinates are compared with the supplied observations.

The L1 error is defined as:

$$L_1 = \sum_{i=1}^N (|x_i - \hat{x}_i| + |y_i - \hat{y}_i|) \quad (39)$$

where:

- x_i, y_i are the observed coordinates,
- $\hat{x}_i, \hat{y}_i$ are the reconstructed coordinates,
- N is the total number of observations.

For the estimated parameters, the total L1 error is approximately:

$$L_1 \approx 0.0308 \quad (40)$$

for 1,500 supplied points.

The corresponding average combined absolute error per point is:

$$\text{Mean L1 Error} = \frac{0.0308}{1500} \quad (41)$$

which gives approximately:

$$\text{Mean L1 Error} \approx 2.06 \times 10^{-5} \quad (42)$$

The very small error indicates that the estimated parameters provide a highly accurate reconstruction of the supplied curve.

12 Validation

The final solution is validated using both numerical and mathematical checks.

12.1 Parameter Constraint Validation

The estimated values are:

$$\theta = 30^\circ \quad (43)$$

$$M = 0.03 \quad (44)$$

$$X = 55 \quad (45)$$

These satisfy:

$$0 < 30 < 50 \quad (46)$$

$$-0.05 < 0.03 < 0.05 \quad (47)$$

and:

$$0 < 55 < 100 \quad (48)$$

Therefore, all parameter constraints are satisfied.

12.2 Hidden Parameter Validation

For each observed point, the corresponding value of t is recovered using:

$$t_i = (x_i - X) \cos(\theta) + (y_i - 42) \sin(\theta) \quad (49)$$

The recovered values are checked to ensure consistency with:

$$6 < t_i < 60 \quad (50)$$

This provides an additional validation of the estimated orientation and horizontal translation parameters.

12.3 Curve Reconstruction Validation

The estimated parameters are substituted back into the original two-dimensional equations rather than evaluating only the transformed optimization objective.

The reconstructed (x, y) coordinates are then compared against the original observations.

This ensures that the final solution is valid in the original problem space.

13 Actual vs Predicted Curve

A visual comparison between the original observations and the reconstructed curve provides an intuitive validation of the fitted parameters.

The expected plot contains:

- Original observed points.

- Reconstructed parametric curve.
- Coordinate axes with appropriate labels.
- Legend identifying observed and predicted data.

If the generated plot is stored as:

`results/fitted_curve.png`

it can be included using the following code:

```
\begin{figure}[H]
  \centering
  \includegraphics[width=0.85\textwidth]
  {results/fitted_curve.png}
  \caption{Actual Observations and Reconstructed Curve}
  \label{fig:fitted_curve}
\end{figure}
```

The fitted curve should closely overlap the observed data points, which is consistent with the very small calculated L1 error.

14 Complete Project Workflow

The complete solution follows the workflow below:

1. Load the x and y observations from the CSV file.
2. Define the mathematical transformation of the parametric curve.
3. Use the candidate θ and X values to recover t .
4. Calculate the transformed oscillatory component.
5. Define the nonlinear residual function.
6. Apply bounded nonlinear least-squares optimization.
7. Estimate θ , M , and X .
8. Substitute the estimated parameters into the original equations.
9. Generate the reconstructed curve.
10. Calculate the L1 reconstruction error.
11. Validate the parameter constraints and recovered t values.
12. Compare the actual and predicted curves visually.

15 Project Architecture

The implementation is organized into separate components to keep the project modular and maintainable.

curve-parameter-estimation/

data/
xy_data.csv

src/
curve_model.py
optimizer.py
evaluation.py

notebooks/
curve_analysis.ipynb

results/
fitted_curve.png
parameters.txt

main.py

requirements.txt

README.md

.gitignore

16 Technologies Used

The project uses the following technologies:

Technology	Purpose
Python	Main programming language
NumPy	Numerical and mathematical computation
Pandas	Dataset loading and processing
SciPy	Nonlinear least-squares optimization
Matplotlib	Curve visualization

Table 3: Technologies Used in the Project

17 Installation and Execution

The project can be executed using a standard Python environment.

The required dependencies are:

```
numpy  
pandas  
scipy  
matplotlib
```

These dependencies can be installed using:

```
pip install -r requirements.txt
```

After installation, the project can be executed from the project root directory using:

```
python main.py
```

The program loads the dataset, performs parameter estimation, reconstructs the curve, calculates the error, and generates the visualization.

18 Results Summary

The final results obtained from the parameter estimation process are summarized below.

Metric / Parameter	Value
Number of observations	1,500
θ	30°
M	0.03
X	55
Total L1 Error	≈ 0.0308
Mean L1 Error	$\approx 2.06 \times 10^{-5}$

Table 4: Final Results

19 Discussion

The parameter estimation problem is nonlinear because the unknown parameters appear inside trigonometric and exponential expressions.

A direct optimization approach could treat both the unknown parameters and the hidden t values as optimization variables. However, this would introduce a large number of additional variables because the dataset contains 1,500 observations.

The mathematical transformation used in this project avoids this high-dimensional optimization.

For a candidate θ and X , the hidden parameter t can be recovered directly using a projection equation. The perpendicular projection then isolates the oscillatory component.

As a result, the optimization is reduced to only three unknown parameters:

$$\theta, \quad M, \quad X \quad (51)$$

This makes the optimization computationally simpler while preserving the structure of the original model.

The final parameter estimates produce a very small reconstruction error, indicating that the recovered parameters accurately describe the supplied dataset.

20 Conclusion

This project demonstrates the application of mathematical analysis and numerical optimization to a nonlinear inverse parameter estimation problem.

The original two-dimensional parametric curve was transformed using orthogonal projections. This transformation allowed the hidden parameter t and the oscillatory component to be recovered for candidate parameter values.

A bounded nonlinear least-squares optimization procedure was then used to estimate the three unknown parameters.

The final estimated parameters are:

$$\boxed{\theta = 30^\circ, \quad M = 0.03, \quad X = 55} \quad (52)$$

The estimated parameters satisfy all specified constraints and produce a highly accurate reconstruction of the observed data.

The total L1 error is approximately:

$$\boxed{L_1 \approx 0.0308} \quad (53)$$

across 1,500 observations.

Therefore, the proposed mathematical transformation combined with bounded nonlinear optimization provides an effective and efficient solution to the parameter recovery problem.

Final Answer

The unknown parameters of the supplied parametric curve are:

$$\boxed{\theta = 30^\circ} \quad (54)$$

$$\boxed{M = 0.03} \quad (55)$$

$$\boxed{X = 55} \quad (56)$$

These values provide a close match to the supplied 1,500-point dataset and satisfy all the parameter constraints specified in the assignment.

— End of Report —